

Large N expansion of QED: asymptotic photon propagator and contributions to the muon anomaly, for any number of loops

D.J. Broadhurst*

Physics Department, Open University, Milton Keynes, MK7 6AA, UK

Received 23 December 1992

Abstract. We calculate analytical contributions to the n -loop asymptotic photon propagator from diagrams with $n-1$ electron loops, i.e. the $O(1/N)$ terms in the large N limit. The corresponding contributions to the on-shell β -function, $\beta(\alpha) = \partial \log \alpha / \partial \log m$ are reduced to rational combinations of $\zeta_s \equiv \sum_p p^{-s}$. For the β -function of the MOM scheme (i.e. the Gell-Man-Low function) we obtain the $O(1/N)$ terms of

$$\Psi(\alpha_{\text{MOM}}) = \beta(\alpha) \partial \log \alpha_{\text{MOM}} / \partial \log \alpha = \sum_n \Psi_n(\alpha_{\text{MOM}}/\pi)^n$$

in closed form:

$$\Psi_n^{[n-1]} = \frac{(n-1)!}{(-3)^{n-1}} \left[-2n + 4 - \frac{n+4}{2^n} + \frac{16}{n-1} \sum_{n/2 > s > 0} s(1-2^{-2s})(1-2^{2s-n}) \zeta_{2s+1} \right].$$

Neglecting terms of order m_e/m_μ , we calculate $(n+1)$ -loop muon-anomaly contributions, obtained by inserting, in the one-loop diagram, all n -loop photon-propagator diagrams with at least $(n-1)$ electron loops. The methods are applicable to any number of loops. Results have been obtained analytically up to 20 loops and numerically up to 100 loops.

1 Introduction

Recent work [1, 2] has yielded analytical 4-loop results for contributions to the asymptotic muon anomaly a_μ^∞ , i.e. to $(g/2-1)_\mu$ in the limit $m_\mu/m_e \rightarrow \infty$. In particular, one now knows analytically the contributions obtained by inserting, in the one-loop muon-anomaly diagram, all three-loop photon propagator diagrams with two electron loops, thanks to a calculation [1] of the corresponding on-shell asymptotic photon self-energy coefficient $a_3^{[2]}$, i.e. the leading three-loop non-logarithmic

term in the large N limit [3], where N is the (fictive) number of electrons. Whilst one evaluates all diagrams with $N=1$, the large N limit has the virtue of picking out gauge-invariant sets of diagrams that give high powers of $\log(m_\mu/m_e)$ in the muon anomaly. Here we shall be concerned with a class of $O(1/N^2)$ contributions to the muon anomaly, for any number of loops. The subject of $O(1/N^3)$ contributions at 4 loops is under intensive investigation in [4, 5], where discrepancies between analytical [6, 7] and numerical approaches [8] are being addressed.

The computation [9] of 5-loop anomaly contributions of the form of Fig. 1 was frustrated by the absence of an analytical or numerical value for $a_4^{[3]}$. Here we remedy that state of affairs and show how to solve the general n -loop problem, reducing it to algebra that may be performed for any value of n , given enough computing power. In practice, we find that analytical computations take minutes of CPUtime for 10 loops and a hour for 20 loops. No integration is required: differentiation and recurrence relations suffice.

Our on-shell β -function is normalized as in [1, 10]:

$$\beta(\alpha) = \frac{\partial \log \alpha}{\partial \log m} = \sum_{n=1}^{\infty} \beta_n(\alpha/\pi)^n. \quad (1)$$

We require the coefficients $\beta_n^{[n-1]}$, where the subscript n denotes the total number of loops in the corresponding diagrams for the photon self energy and the superscript $[n-1]$ indicates the number of electron loops. Approx-

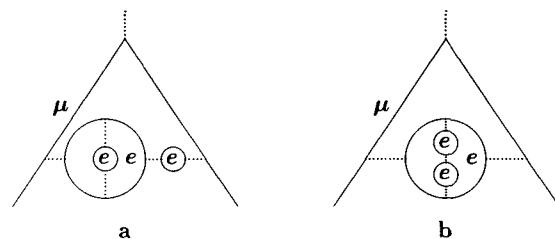


Fig. 1. Examples of 5-loop $O(1/N^2)$ muon-anomaly contributions

* d. broadhurst@open.ac.uk

mate numerical values up to $n=10$ were given in [11]. We shall obtain analytical results in the form of linear combinations of $\zeta_s \equiv \sum_p p^{-s}$, with rational coefficients. We also require the corresponding coefficients $a_n^{l^{n-1}}$ of the non-logarithmic terms in the asymptotic on-shell photon self energy $\Pi(L, \alpha) = \sum_n \Pi_n(L) (\alpha/\pi)^n$, whose n -loop coefficient $\Pi_n(L)$ is a polynomial in $L \equiv \log(-q^2/m_e^2)$, with $\Pi_n(0) = a_n$. The dependence on the logarithm L is given by [10]

$$\Pi(L, \alpha) = -1 + \exp \left[\frac{1}{2} L \beta(\alpha) \left(\alpha \frac{\partial}{\partial \alpha} - 1 \right) \right] \left(1 + \sum_{n=1}^{\infty} a_n (\alpha/\pi)^n \right), \quad (2)$$

which is the solution to the Callan-Symanzik equation for the scheme-invariant coupling

$$\hat{\alpha}(L, \alpha) = \frac{\alpha}{1 + \Pi(L, \alpha)} = \frac{\bar{\alpha}}{1 + \bar{\Pi}(\bar{L}, \bar{\alpha})}, \quad (3)$$

where the final form is the corresponding $\overline{\text{MS}}$ expression, with $\bar{L} \equiv \log(-q^2/\mu^2)$, for a $\overline{\text{MS}}$ coupling $\bar{\alpha}$ renormalized at a scale μ .

To obtain $a_n^{l^{n-1}}$, we find it convenient to compute the corresponding quantity $\bar{a}_n^{l^{n-1}}$ in the $\overline{\text{MS}}$ scheme. A method for obtaining the coefficients $\bar{\beta}_n^{l^{n-1}}$ of the $\overline{\text{MS}}$ β -function, $\bar{\beta}(\bar{\alpha}) = \partial \log \bar{\alpha} / \partial \log \mu$, was given in [3], where an integral representation of the $O(1/N)$ term in $\bar{\beta}$ was found. We shall extend this method to compute $\bar{a}_n^{l^{n-1}}$, and hence the coefficients of the β -function in the MOM scheme, i.e. the Gell-Mann-Low function:

$$\begin{aligned} \Psi(\alpha_{\text{MOM}}) &= \sum_{n=1}^{\infty} \Psi_n(\alpha_{\text{MOM}}/\pi)^n = \beta(\alpha) \frac{\partial \log \alpha_{\text{MOM}}}{\partial \log \alpha} \\ &= \bar{\beta}(\bar{\alpha}) \frac{\partial \log \alpha_{\text{MOM}}}{\partial \log \bar{\alpha}}, \end{aligned} \quad (4)$$

where α_{MOM} is the value of the scheme-invariant coupling (3) at some (arbitrarily chosen) large value of $-q^2$. From Ψ and the β -function of a given scheme, one may construct the asymptotic photon propagator in that scheme. In particular, we find that

$$\begin{aligned} (n-1) \beta_1^{l^{11}} \Pi_n^{l^{n-1}}(L) \\ = \beta_{n+1}^{l^{n1}} - \sum_{p=1}^n \binom{n-1}{p-1} \Psi_{p+1}^{l^{p1}} [-\Pi_1^{l^{11}}(L)]^{n-p}, \end{aligned} \quad (5)$$

where $\beta_1^{l^{11}} = \frac{2}{3}$ and $\Pi_1^{l^{11}}(L) = \frac{5}{9} - \frac{1}{3}L$.

Finally, we shall expand the coupling (3) in powers of α and $L = \log(-q^2/m_e^2)$ and hence calculate the $(n+1)$ -loop, $(n-1)$ -electron-loop contributions to a_n^{∞} , in terms of $\log(m_\mu/m_e)$ and $\{\zeta_s | s \leq n+1\}$, by finding analytical expressions for the integrals [10]

$$I_n \equiv \int_0^1 dy (1-y) \log^n \left(\frac{y^2}{1-y} \right), \quad (6)$$

which arise from integration over $-q^2 = y^2 m_\mu^2 / (1-y)$.

2 Gell-Mann-Low function

First we derive the $\overline{\text{MS}}$ results for $\bar{a}_n^{l^{n-1}}$ by extending the method used in [3] for $\bar{\beta}_n^{l^{n-1}}$. Let the n -loop, $(n-1)$ -electron-loop coefficient of $(\alpha_0/\pi)^n$ in the bare photon self energy be expressed in the form [3]

$$\Pi_{0,n}^{l^{n-1}} = \left(-\frac{1}{3\varepsilon} \right)^{n-2} \frac{F(\varepsilon, n\varepsilon)}{4n\varepsilon}, \quad (7)$$

where $\alpha_0 = (4\pi/\mu^2 e^\gamma) e_0^2 / 4\pi$, with e_0 the bare charge in $d = 4 - 2\varepsilon$ spacetime dimensions, and μ the $\overline{\text{MS}}$ mass unit, introduced to make α_0 dimensionless, in such a way as to suppress terms involving $\log 4\pi$ and Euler's constant, γ . Then $F(\varepsilon, \delta)$ is analytic in the neighbourhood of $\varepsilon = \delta = 0$, where it may be expanded as a double power series

$$F(\varepsilon, \delta) = \sum_{n,m \geq 0} F_{n,m} \varepsilon^n \delta^m. \quad (8)$$

We have proved the validity of expansion (8) by reducing F to Γ -functions and a single Saalschützian F_{32} series, of the form studied intensively in [7]. The explicit form of this reduction will not be needed here. We now transform the series in the bare coupling to one in the $\overline{\text{MS}}$ coupling, using the one-loop charge renormalization $\alpha_0 = \bar{\alpha}/(1 - \bar{\alpha}/3\pi\varepsilon)$ and the double expansion (8). The resultant quadruple series is vastly simplified by combinatoric identities [3], enabling one to isolate the term involving $\bar{\alpha}^n/\varepsilon$ that determines $\bar{\beta}_n^{l^{n-1}}$. We have extended the analysis of [3] to identify the finite terms that survive as $\varepsilon \rightarrow 0$. The results are much simpler than the method:

$$2(n-1) 3^{n-2} \bar{\beta}_n^{l^{n-1}} = F_{n-2,0}, \quad (9)$$

$$4n(n-1) 3^{n-2} \bar{\Pi}_n^{l^{n-1}} = F_{n-1,0} + n! (-1)^n F_{0,n-1}. \quad (10)$$

The first result is that given in [3], for which one requires only the expansion of

$$F(\varepsilon, 0) = \frac{(1+\varepsilon)(1-2\varepsilon)(3-2\varepsilon)}{54B(2-\varepsilon, 2-\varepsilon)B(3-\varepsilon, 1+\varepsilon)} \quad (11)$$

to determine $\bar{\beta}_n^{l^{n-1}}$. The second is new and shows that one also needs the expansion of $F(0, \delta)$ to determine $\bar{a}_n^{l^{n-1}}$. Systematic application of the F_{32} identities of [7] gives

$$\begin{aligned} F(0, \delta) &= (-\mu^2 e^{5/3}/q^2)^\delta P(1-\delta), \\ P(x) &= \frac{32}{3(1+x)} \sum_{k=2}^{\infty} \frac{(-1)^k k}{(k^2 - x^2)^2}. \end{aligned} \quad (12)$$

Note that the F_{32} hypergeometric series that complicates (7) has been reduced in (12) to a trigamma function of the type previously encountered in [12] and exploited in [13]. It is akin to trigamma functions found in the large N expansion of critical exponents [14, 15].

Expanding the Euler Beta-functions of (11), we reproduce the results for $\{\bar{\beta}_n^{l^{n-1}} | n \leq 8\}$ of (4.2) of [3], with the exception of the sign of the ζ_3^2 term in $\bar{\beta}_8^{l^{71}}$, which

Table 1. Analytical contributions to Ψ , β , Π and a_μ^∞ .

$$\begin{aligned} \sum_{n \leq 10} \Psi_n^{[n-1]} x^n &= \frac{1}{2} x^2 + \left[-\frac{23}{36} + \frac{2}{3} \zeta_3 \right] x^3 + \left[1 - \frac{2}{3} \zeta_3 \right] x^4 + \left[-\frac{67}{36} + \frac{7}{9} \zeta_3 + \frac{10}{9} \zeta_5 \right] x^5 \\ &+ \left[\frac{145}{36} - \frac{10}{9} \zeta_3 - \frac{20}{9} \zeta_5 \right] x^6 + \left[-\frac{6455}{648} + \frac{155}{81} \zeta_3 + \frac{350}{81} \zeta_5 + \frac{35}{9} \zeta_7 \right] x^7 + \left[\frac{8995}{324} - \frac{35}{9} \zeta_3 - \frac{250}{27} \zeta_5 - \frac{35}{3} \zeta_7 \right] x^8 \\ &+ \left[-\frac{251335}{2916} + \frac{4445}{486} \zeta_3 + \frac{5425}{243} \zeta_5 + \frac{1715}{54} \zeta_7 + \frac{5950}{243} \zeta_9 \right] x^9 \\ &+ \left[\frac{31885}{108} - \frac{5950}{243} \zeta_3 - \frac{4900}{81} \zeta_5 - \frac{2450}{27} \zeta_7 - \frac{23800}{243} \zeta_9 \right] x^{10} \end{aligned}$$

$$\begin{aligned} \sum_{n \leq 10} \beta_n^{[n-1]} x^n &= \frac{1}{2} x^2 - \frac{7}{9} x^3 + \left[\frac{901}{648} - \frac{8}{9} \zeta_2 - \frac{7}{48} \zeta_3 \right] x^4 + \left[-\frac{203393}{102060} + \frac{32}{135} \zeta_2 + \frac{493}{420} \zeta_3 \right] x^5 \\ &+ \left[\frac{4078741}{1312200} - \frac{6376}{2025} \zeta_3 - \frac{64}{27} \zeta_4 + \frac{93}{32} \zeta_5 \right] x^6 + \left[-\frac{1274016383}{236432196} + \frac{4640}{729} \zeta_3 + \frac{18080}{56133} \zeta_4 - \frac{2788355}{972972} \zeta_5 \right] x^7 \\ &+ \left[\frac{11172347027}{1041624360} - \frac{24320}{2187} \zeta_3 + \frac{837376}{535815} \zeta_4 + \frac{141298}{11907} \zeta_5 - \frac{320}{27} \zeta_6 \right] x^8 \\ &+ \left[-\frac{38225950250339281}{1546443885987000} + \frac{336896}{19683} \zeta_3 - \frac{8704}{18225} \zeta_4 - \frac{40431728}{1148175} \zeta_5 - \frac{7797440}{5316597} \zeta_6 + \frac{50770048}{1247103} \zeta_7 \right] x^9 \\ &+ \left[\frac{47037454886664856}{719445228913125} - \frac{12845056}{531441} \zeta_3 + \frac{28672}{820125} \zeta_4 + \frac{16974848}{273375} \zeta_5 + \frac{1351238912}{59108049} \zeta_6 - \frac{77160194944}{2631715515} \zeta_7 \right. \\ &\left. - \frac{71680}{729} \zeta_8 \right] x^{10} \end{aligned}$$

$$\begin{aligned} \sum_{n \leq 5} a_n^{[n-1]} x^n &= \left[\frac{5}{24} - \zeta_3 \right] x^2 + \left[-\frac{307}{864} - \frac{2}{3} \zeta_2 + \frac{545}{576} \zeta_3 \right] x^3 + \left[\frac{75259}{68040} + \frac{16}{135} \zeta_2 - \frac{15109}{22680} \zeta_3 - \frac{5}{9} \zeta_5 \right] x^4 \\ &+ \left[-\frac{9701009}{3499200} + \frac{37799}{72900} \zeta_3 - \frac{8}{9} \zeta_4 + \frac{19693}{6912} \zeta_5 \right] x^5 \end{aligned}$$

$$\begin{aligned} \sum_{n \leq 6} A_n^{[n-1]} x^n &= \frac{1}{2} x + \left[-\frac{25}{36} + \frac{1}{3} l \right] x^2 + \left[\frac{317}{324} + \frac{2}{9} \zeta_2 - \frac{25}{27} l + \frac{2}{9} l^2 \right] x^3 \\ &+ \left[-\frac{8609}{5832} - \frac{25}{27} \zeta_2 - \frac{2}{9} \zeta_3 + \left(\frac{317}{162} + \frac{4}{9} \zeta_2 \right) l - \frac{25}{27} l^2 + \frac{4}{27} l^3 \right] x^4 \\ &+ \left[\frac{64613}{26244} + \frac{634}{243} \zeta_2 + \frac{100}{81} \zeta_3 + \frac{4}{3} \zeta_4 + \left(-\frac{8609}{2187} - \frac{200}{81} \zeta_2 - \frac{16}{27} \zeta_3 \right) l + \left(\frac{634}{243} + \frac{16}{27} \zeta_2 \right) l^2 - \frac{200}{243} l^3 + \frac{8}{81} l^4 \right] x^5 \\ &+ \left[-\frac{2182775}{472392} - \frac{43045}{6561} \zeta_2 - \frac{3170}{729} \zeta_3 - \frac{250}{27} \zeta_4 - \frac{40}{27} \zeta_5 - \frac{80}{81} \zeta_2 \zeta_3 + \left(\frac{323065}{39366} + \frac{6340}{729} \zeta_2 + \frac{1000}{243} \zeta_3 + \frac{40}{9} \zeta_4 \right) l \right. \\ &\left. + \left(-\frac{43045}{6561} - \frac{1000}{243} \zeta_2 - \frac{80}{81} \zeta_3 \right) l^2 + \left(\frac{6340}{2187} + \frac{160}{243} \zeta_2 \right) l^3 - \frac{500}{729} l^4 + \frac{16}{243} l^5 \right] x^6 \end{aligned}$$

$$\begin{aligned} \sum_{n \leq 6} A_n^{[n-2]} x^n &= \left[-\frac{5}{12} + \frac{1}{2} \zeta_3 + \frac{1}{4} l \right] x^3 + \left[\frac{1189}{576} + \frac{3}{4} \zeta_2 - \frac{875}{384} \zeta_3 + \left(-\frac{23}{12} + \zeta_3 \right) l + \frac{5}{12} l^2 \right] x^4 \\ &+ \left[-\frac{1379117}{181440} - \frac{809}{180} \zeta_2 + \frac{760369}{120960} \zeta_3 + \frac{5}{18} \zeta_5 + \frac{4}{3} \zeta_2 \zeta_3 + \left(\frac{3779}{432} + \frac{17}{9} \zeta_2 - \frac{1771}{288} \zeta_3 \right) l + \left(-\frac{379}{108} + \frac{4}{3} \zeta_3 \right) l^2 \right. \\ &\left. + \frac{13}{27} l^3 \right] x^5 + \left[\frac{38889259}{1530900} + \frac{2909}{144} \zeta_2 - \frac{406665001}{32659200} \zeta_3 + \frac{287}{36} \zeta_4 - \frac{119879}{41472} \zeta_5 - \frac{8897}{864} \zeta_2 \zeta_3 - \frac{20}{9} \zeta_3^2 \right. \\ &\left. + \left(-\frac{27351557}{816480} - \frac{12959}{810} \zeta_2 + \frac{3864443}{181440} \zeta_3 + \frac{25}{27} \zeta_5 + \frac{40}{9} \zeta_2 \zeta_3 \right) l + \left(\frac{70295}{3888} + \frac{89}{27} \zeta_2 - \frac{8897}{864} \zeta_3 \right) l^2 \right. \\ &\left. + \left(-\frac{2279}{486} + \frac{40}{27} \zeta_3 \right) l^3 + \frac{77}{162} l^4 \right] x^6 \end{aligned}$$

is misprinted in [3]. The quantity of significance to our on-shell calculation is the scheme-invariant Gell-Mann-Low coefficient $\Psi_n^{[n-1]}$, which is most simply found by choosing the MOM-scheme euclidean point, at which the self energy vanishes, to be $q^2 = -\mu^2 e^{5/3}$, so that $\alpha_{\text{MOM}} = \bar{\alpha}$ to leading order in $1/N$. Thus we obtain a result

$$\begin{aligned} \Psi_n^{[n-1]} &= \frac{1}{2} \left(\frac{1}{3} \frac{d}{dx} \right)^{n-2} P(x) \Big|_{x=1} \\ &= \frac{(n-1)!}{(-3)^{n-1}} \left[-2n + 4 - \frac{n+4}{2^n} \right. \\ &\quad \left. + \frac{16}{n-1} \sum_{n/2 > s > 0} s(1-2^{-2s})(1-2^{2s-n}) \zeta_{2s+1} \right], \quad (13) \end{aligned}$$

Table 2. Numerical values of coefficients, obtained from analytical results

n	$\psi_n^{[n-1]}$	$\beta_n^{[n-1]}$	$a_n^{[n-1]}$	$A_n^{[n-1]}$	$A_n^{[n-2]}$
2	0.5	0.5	-0.9937236	1.0827551	0
3	0.1624824	-0.7777778	-0.3145839	2.7241470	1.5172614
4	0.1986287	-0.2470315	-0.0758047	7.1966590	8.5930057
5	0.2259640	-0.1919806	-0.1568294	20.183187	38.058186
6	0.3878751	-0.2284598	-0.1294930	58.186078	154.76338
7	0.7407229	-0.3605866	-0.3058231	175.51860	604.65837
8	1.7224221	-0.7022122	-0.5273543	535.57609	2339.2810
9	4.5111386	-1.6170472	-1.4069621	1721.9937	8954.5365
10	13.433219	-4.2839482	-3.6784392	5416.3787	34814.141
11	44.355375	-12.809843	-11.726336	18989.508	133625.08
12	161.61002	-42.629609	-39.808543	56744.572	540262.60
13	642.51848	-156.18043	-151.71184	254768.23	2042188.7
14	2770.7644	-624.46515	-623.54140	396651.98	9368299.8
15	12874.061	-2705.4589	-2788.6110	6026975.0	28832707.
16	64131.438	-12624.173	-13384.421	-23330213.	2.2755261×10^8
17	340906.24	-63117.610	-68862.370	3.9144940×10^8	-1.4266218×10^8
18	1926191.8	-336618.55	-377387.55	-3.8766338×10^9	1.2667419×10^{10}
19	11527126.	-1907480.5	-2196543.2	4.8755448×10^{10}	$-1.0366284 \times 10^{11}$
20	72835249.	-11444809.	-13525394.	$-6.0847944 \times 10^{11}$	1.6140381×10^{12}

from which the ζ -products and ζ_{2s} terms of the $\overline{\text{MS}}$ β -function are absent. It may be possible (and indeed easier) to obtain this result by scaling arguments at the non-trivial fixed-point in d dimensions, which occurs for a coupling of order ε in the large N limit [16].

The entries in Table 1 for $\Psi_n^{[n-1]}$ with $n \leq 4$ agree with previous results [17]. The rest are new. For large n , we find the behaviour

$$\frac{\Psi_n^{[n-1]}}{(n-2)!} = \frac{6n-1}{3^{n+1}} - \frac{6n+1}{6^{n+1}} - \frac{3}{(-6)^{n-1}} - \frac{6nK_1(n)}{(-9)^n}, \quad (14)$$

where $K_1(n) = 1 \mp \frac{1}{10} + \frac{7}{2n} \left(1 \mp \frac{1}{100} \right) + O((3/4)^n)$, for even or odd n , and $|K_1(n) - 1| < 1$ for all $n > 3$. The constant-sign factorial growth, exhibited in Table 2, is to be expected: the perturbation series is not Borel summable for $\alpha > 0$, because of the Landau singularity associated with the one-loop β -function, which dominates as $N \rightarrow \infty$.

3 On-shell β function

Now we compute the coefficients $\beta_n^{[n-1]}$ of the on-shell β -function. The results [10] $\beta_2^{[1]} = \frac{1}{2}$ and $\beta_3^{[2]} = -\frac{7}{9}$ are easily obtained from the expression [11, 18]

$$\beta_n^{[n-1]} = - \int_0^1 d\theta [-\omega_\infty]^{n-2} \frac{dG}{d\theta}, \quad (15)$$

where $\omega_\infty = \frac{5}{9} - \frac{1}{3} \log(-q^2/m^2)$ is the asymptotic form of the one-loop self energy, θ is an integration variable, in terms of which $-q^2 = m^2(1-\theta)^2/\theta$, and the weight function is the differential of $G(\theta) = \frac{1}{8}\Xi(\theta) + (1-\theta)(1-\frac{1}{3}\theta)$, where the ‘direct’ term involves a one-loop light-by-light scattering function, Ξ , that is known analytically [18] and the second contribution is a counterterm, from mass renormalization.

The important observation of [1, 11] is that (15) does not give $\beta_n^{[n-1]}$ for $n > 3$, which means that the 4-loop analytical calculation of [19] and the higher-loop numerical calculations of [18] need correction. The correction term is given by [11]

$$\beta_n^{[n-1]} - \tilde{\beta}_n^{[n-1]} = -\frac{n-2}{3} \int_0^1 \frac{d\theta}{\theta} ([-\omega]^{n-3} - [-\omega_\infty]^{n-3}) \frac{1+\theta}{1-\theta} G, \quad (16)$$

where

$$\omega(\theta) = \frac{5}{9} - \frac{4\theta(1-\theta) - (1-4\theta+\theta^2)(1+\theta)\log\theta}{3(1-\theta)^3} \quad (17)$$

is the *exact* on-shell one-loop self energy. Evaluating the integral (16) for $n=4$, the authors of [1, 11] resolved a discrepancy between analytical [2] and numerical [8] calculations of 4-loop $O(1/N^2)$ muon-anomaly contributions. In [11] the coefficients $\{\beta_n^{[n-1]} | n=5, 10\}$ were given to two significant figures. We shall obtain them analytically.

The integrals of (15) and (16) may be combined to give the more transparent result

$$\beta_n^{[n-1]} = - \int_0^1 d\theta \left(\frac{n-2}{3\theta} [-\omega]^{n-3} \frac{1+\theta}{1-\theta} G + \frac{d}{d\theta} ([-\omega_\infty]^{n-2} G) \right), \quad (18)$$

which makes it clear that the unphysical asymptotic approximation, ω_∞ , enters the integrand only in a total differential, serving to subtract off the non-integrable behaviour of the physical term at $\theta=0$. For our purposes, it is best to replace ω_∞ by ω , and $G(\theta)$ by $G(0) = \frac{1}{2}$, in the differential. One may do this with impunity for

all $n > 2$, since our new regulator, $[-\omega]^{n-2} G(0)$, has the same endpoint behaviour as the regulator $[-\omega_\infty]^{n-2} G(\theta)$ that results from the analysis of [11]. Taking into account the different $\theta \rightarrow 1$ behaviour in the special case $n=2$, we obtain

$$\beta_n^{[n-1]} = \frac{1}{2} \delta_{n,2} - \frac{n-2}{3} \int_0^1 d\theta [-\omega]^{n-3} \left\{ \frac{1+\theta}{1-\theta} \frac{G}{\theta} - \frac{3}{2} \frac{d\omega}{d\theta} \right\}. \quad (19)$$

The final step is to observe that the combination of functions in braces is the differential of an elementary function, $N_1 + N_2$, consisting of direct and counterterm contributions:

$$N_1(\theta) = -\frac{2\theta^2(3+2\theta+3\theta^2)\log\theta}{(1-\theta^2)^3} - \frac{\theta(1+6\theta+\theta^2)}{(1-\theta^2)^2}, \quad (20)$$

$$N_2(\theta) = \frac{2\theta^2(3-\theta)\log\theta}{(1-\theta)^3} + \frac{\theta(28-9\theta+6\theta^2-\theta^3)}{6(1-\theta)^2}, \quad (21)$$

in terms of which we obtain our final expression

$$\beta_n^{[n-1]} = \frac{1}{2} \delta_{n,2} - \frac{7}{9} \delta_{n,3} + \frac{n-2}{3} \int_0^1 d\theta (N_1 + N_2) \frac{d}{d\theta} [-\omega]^{n-3}, \quad (22)$$

from integrating (19) by parts and taking into account the values $N_1(1) = \frac{1}{6}$ and $N_2(1) = \frac{13}{6}$, in the special case $n=3$.

Much is achieved by transforming integrals (15) and (16) to give (22). Firstly, the two-loop and three-loop results of [10] are immediate consequences of the values $G(0) = \frac{1}{2}$ and $N_1(1) + N_2(1) = \frac{7}{3}$. Secondly, the large n behaviour, undetermined in [11], is easily found by expanding $N_{1,2}$ in powers of ω and $\exp(3\omega - 5/3) \approx \theta$, in the

limit $\theta \rightarrow 0$, and using ω as the integration variable in (22), which gives

$$\frac{\beta_n^{[n-1]}}{(n-2)!} = -\frac{11}{3^{n-1}e^{5/3}} - \frac{33}{6^{n-2}e^{10/3}} + \frac{34n-355}{9^{n-2}e^5} + \frac{16(437nK_2(n)-2999)}{12^{n-1}e^{20/3}}, \quad (23)$$

where $K_2(n) = 1 + O(n(4/5)^n)$ and $|K_2(n) - 1| < 1$ for all $n > 2$. Finally, the integral can be reduced to ζ -functions by purely algebraic methods, as will now be shown.

In (22), we make the transformations $\theta = (1-x_1)/(1+x_1)$ and $\theta = 1-x_2$, for the $N_{1,2}$ terms, respectively, and find that the denominator of each integrand is purely a power of $x_{1,2}$, respectively. As no non-trivial partial fractions occur, the coefficients $\beta_n^{[n-1]}$ are given algebraically as rational combinations of integrals of the form

$$I_n^m = \frac{(-\frac{1}{2})^m}{m!} \int_{\varepsilon_1}^1 \frac{dx_1}{x_1^n} \log^m \theta, \quad J_n^m = \frac{(-1)^m}{m!} \int_{\varepsilon_2}^1 \frac{dx_2}{x_2^n} \log^m \theta, \quad (24)$$

where only the finite terms in the limits $\varepsilon_{1,2} \rightarrow 0$ are to be retained, since the powers of $1/\varepsilon_{1,2}$ cancel for the combinations of integrals in question. Moreover, the direct integrals, I_n^m , have $n+m$ odd. The problem is now reduced to algebra by the following relations, obtained from integration by parts:

$$\begin{aligned} (n-1) I_n^m &= (n-3) I_{n-2}^m + I_{n-1}^{m-1} + K_{n-1}^m - K_{n-3}^m, \\ I_1^{2s} &= I_0^{2s+1} + 2^{-2s} \zeta_{2s+1}, \\ (n-1) J_n^m &= (n-2) J_{n-1}^m + J_{n-1}^{m-1} + L_{n-1}^m - L_{n-2}^m, \\ J_1^s &= \zeta_{s+1}, \\ n K_n^m &= (n-2) K_{n-2}^m + K_{n-1}^{m-1}, \quad K_0^s = \delta_{s,0}, \\ n L_n^m &= (n-1) L_{n-1}^m + L_{n-1}^{m-1}, \quad L_0^s = \delta_{s,0}, \\ I_0^{2s+1} &= (2^{-2s} - 2^{-4s}) \zeta_{2s+1}, \end{aligned} \quad (25)$$

Table 3. Asymptotic behaviour: $K_{1,2}(n)$ are defined in (14) and (23); $A(n) \equiv (2/3)^n n!$; $\bar{A}_n$ and $\hat{A}_n$ are the coefficients of (26) in the $\overline{\text{MS}}$ and MOM schemes, renormalized at m_μ

n	$K_1(n)$	$K_2(n)$	$\frac{\alpha_n^{[n-1]}}{\beta_n^{[n-1]}}$	$\frac{A_{n+1}^{[n]}}{A(n)}$	$\frac{\bar{A}_{n+1}^{[n]}}{A(n)}$	$\frac{\hat{A}_{n+1}^{[n]}}{A(n)}$	$\frac{A_{n+1}^{[n-1]}}{A(n)}$	$\frac{\bar{A}_{n+1}^{[n-1]}}{A(n)}$	$\frac{\hat{A}_{n+1}^{[n-1]}}{A(n)}$
10	1.228	1.217	0.859	0.302	2.296	0.999	2.123	-4.382	-2.009
11	1.402	1.171	0.915	0.123	-2.298	-1.000	1.171	4.636	2.109
12	1.179	1.136	0.934	0.069	2.300	1.000	0.553	-4.865	-2.200
13	1.361	1.108	0.971	0.012	-2.300	-1.000	0.293	5.074	2.284
14	1.142	1.086	0.999	0.020	2.301	1.000	0.097	-5.266	-2.362
15	1.330	1.068	1.031	-0.008	-2.301	-1.000	0.076	5.445	2.435
16	1.113	1.054	1.060	0.012	2.301	1.000	-0.004	-5.612	-2.503
17	1.305	1.043	1.091	-0.011	-2.301	-1.000	0.035	5.768	2.567
18	1.091	1.034	1.121	0.011	2.301	1.000	-0.024	-5.916	-2.628
19	1.284	1.027	1.152	-0.011	-2.301	-1.000	0.029	6.056	2.686
20	1.072	1.022	1.182	0.011	2.301	1.000	-0.028	-6.188	-2.741
25	1.241	1.007	1.333	-0.011	-2.301	-1.000	0.032	6.676	2.982
30	1.015	1.002	1.485	0.011	2.301	1.000	-0.034	-7.236	-3.180
35	1.201	1.001	1.636	-0.011	-2.301	-1.000	0.036	7.634	3.348
40	0.987	1.000	1.788	0.011	2.301	1.000	-0.038	-7.980	-3.494
55	1.164	1.000	2.242	-0.011	-2.301	-1.000	0.042	8.803	3.845
70	0.949	1.000	2.697	0.011	2.301	1.000	-0.045	-9.427	-4.112
85	1.142	1.000	3.152	-0.011	-2.301	-1.000	0.048	9.930	4.328
100	0.935	1.000	3.606	0.011	2.301	1.000	-0.050	-10.35	-4.509

where K_n^m and L_n^m are the finite terms in the Laurent expansions of $(\operatorname{arctanh} \varepsilon)^m/m! \varepsilon^n$ and $(-\log(1-\varepsilon))^m/m! \varepsilon^n$, respectively, and hence vanish for $m > n$. We have implemented these recurrence relations as a REDUCE [20] algorithm that reduces $\beta_n^{[n-1]}$ to a rational number plus rational multiples of ζ -functions. It took 4 min of CPU-time to find $\beta_{10}^{[9]}$ and 70 min to find $\beta_{20}^{[19]}$, which contains ζ_{18} and a fraction involving a 60-digit integer. Tables 1 and 2 give a selection of our analytical and numerical results for $\beta_n^{[n-1]}$ and $a_n^{[n-1]}$. The latter is found from (5) and has a similar asymptotic behaviour: $a_n^{[n-1]}/\beta_n^{[n-1]} = (n+19)/33 + O((1/2)^n)$, shown in Table 3.

4 Muon-anomaly contributions

We now calculate contributions to a_μ^∞ that come from insertion of the asymptotic photon propagator in the one-loop muon-anomaly diagram [1, 2, 9]. The sum of these is determined by the asymptotic coupling (3), which gives a total contribution

$$\begin{aligned} a_\mu^\infty [\text{insertion}] &= \int_0^1 dy (1-y) \hat{\alpha}(L, \alpha)/\pi \\ &= \sum_{n=1}^{\infty} A_n (\alpha/\pi)^n \\ &= \sum_{n=1}^{\infty} \bar{A}_n (\bar{\alpha}/\pi)^n = \sum_{n=1}^{\infty} \hat{A}_n (\alpha_{\text{MOM}}/\pi)^n, \end{aligned} \quad (26)$$

where $L \equiv 2l + \log(y^2/(1-y))$ and the dependence on $l \equiv \log(m_\mu/m_e)$ is determined by the renormalization group [21, 22]. Moreover, the leading contributions, at $O(1/N)$ in the large N limit, are easily calculated. The $(n+1)$ -loop, n -electron-loop term is

$$A_{n+1}^{[n]} = \int_0^1 dy (1-y) [-\Pi_1^{[1]}(L)]^n = \sum_{p=0}^n \binom{n}{p} \frac{(6l-5)^p}{3^{n+p}} I_{n-p}, \quad (27)$$

where the integrals (6) are simply obtained by differentiation of a Beta-function:

$$I_n = (d/dx)^n B(2-x, 1+2x)|_{x=0}. \quad (28)$$

On-shell results are presented in Tables 1 and 2. We find that $A_5^{[4]}$ agrees with [9]. Eventually the coefficients $A_{n+1}^{[n]}$ alternate in sign, with magnitudes increasing like $(2/3)^n n! \equiv A(n)$. In the three schemes, one obtains the limits

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{|A_{n+1}^{[n]}|}{A(n)} &= \frac{m_e}{m_\mu} e^{5/6}, \quad \lim_{n \rightarrow \infty} \frac{|\bar{A}_{n+1}^{[n]}|}{\bar{A}(n)} = e^{5/6}, \\ \lim_{n \rightarrow \infty} \frac{|\hat{A}_{n+1}^{[n]}|}{\hat{A}(n)} &= 1, \end{aligned} \quad (29)$$

with the coupling renormalized at m_μ in the $\overline{\text{MS}}$ and MOM schemes. The approach to this asymptotic behaviour is delayed in the on-shell scheme, as can be seen from Table 3.

We now extend the analysis to $O(1/N^2)$, by calculating

$$\begin{aligned} A_{n+1}^{[n-1]} &= - \sum_{p=2}^n (n+1-p) \int_0^1 dy (1-y) \Pi_p^{[p-1]}(L) \\ &\quad \cdot [-\Pi_1^{[1]}(L)]^{n-p}, \end{aligned} \quad (30)$$

where the terms with $p < n$ come from improper terms in the photon propagator, as in Fig. 1a. Now (5) can be used for the $O(1/N)$ self-energy terms and the resulting double sum reduced to a single sum, to give

$$\begin{aligned} \beta_1^{[1]} A_{n+1}^{[n-1]} &= n [\psi(n+1) - \psi(2)] \Psi_2^{[1]} A_n^{[n-1]} \\ &\quad + \sum_{p=2}^n \left[\binom{n}{p} \Psi_{p+1}^{[p]} - (n+1-p) \beta_{p+1}^{[p]} \right] \frac{A_{n+1-p}^{[n-p]}}{p-1}, \end{aligned} \quad (31)$$

where the first term involves $\psi(n+1) = \Gamma'(n+1)/n! = \log(n) + O(1/n)$ and dominates at large n , giving the logarithmic term in the mass-independent asymptotic behaviour

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\frac{8}{9} \frac{A_{n+1}^{[n-1]}}{A_{n+1}^{[n]}} + \log(n) \right) \\ = 1 - \gamma + \int_{1/2}^1 dx \frac{2x-1}{1-x} [1 - P(x)] = 0.6072330782 \dots \end{aligned} \quad (32)$$

Thanks to the renormalization group, the $O(1/N^2)$ terms (31) are linear combinations of the $O(1/N)$ terms (27), with mass-independent coefficients, linear in ζ -functions, coming from the Ψ and β functions at $O(1/N)$. Moreover, the scheme dependence of this relation is negligible at large n , where the Ψ terms dominate the β terms.

Analytical and numerical $O(1/N^2)$ muon-anomaly results are given in Tables 1 and 2. Table 3 reveals that the asymptotic behaviour (32) sets in earlier in the $\overline{\text{MS}}$ and MOM schemes. Previous analytical 4-loop results [1, 2, 11] are verified. We reproduce the combination $A_5^{[3]} + \frac{1}{2} a_4^{[3]}$ given in [9], where neither term was evaluated individually.

5 Summary

From the new results of (13, 22), one may compute, to any order in α , the $O(1/N)$ terms in the asymptotic photon propagator and the corresponding $O(1/N^2)$ muon-anomaly contributions of the type illustrated in Fig. 1.

The $O(1/N)$ coefficients (13) of the Gell-Mann-Low function (4) are given in closed form, for an arbitrary number of loops. Previously, they were known only up to 4 loops [17].

The $O(1/N)$ coefficients (22) of the on-shell β -function are rational linear combinations of ζ -functions, which may be calculated analytically by using the recurrence relations (25) for the finite parts of the integrals (24), generated by the direct and counterterm contributions (20, 21) to the integrand of (22). We have obtained them analytically up to 20 loops and to high numerical preci-

sion up to 100 loops. Previously, they were known analytically up to 4 loops [6] and to low precision up to 10 loops [11].

From the Gell-Mann-Low function, Ψ , and the Callan-Symanzik function, β , one may reconstruct the asymptotic photon propagator [10]. The non-logarithmic coefficients $a_n^{[n-1]}$ have been computed analytically up to 19 loops and to high numerical precision up to 100 loops. Previously, they were unknown beyond 3 loops. Asymptotically, $a_n^{[n-1]}/\beta_n^{[n-1]} \sim (n+19)/33$.

Inserting the leading and $O(1/N)$ photon-propagator terms in the one-loop muon-anomaly diagram, one then obtains the $O(1/N)$ and $O(1/N^2)$ contributions (27) and (31), respectively, to the on-shell coefficients of (26). The $O(1/N^2)$ contributions were previously unknown beyond 4 loops. The dependence on $l = \log(m_\mu/m_e)$ enters solely via the $O(1/N)$ coefficients $A_{n+1}^{[n]}$; it is absorbed into the coupling by renormalizing at a scale m_μ in the $\overline{\text{MS}}$ or MOM schemes. In the $\overline{\text{MS}}$ scheme one sets $l=0$ in (27) and replaces the coefficients of β by those of $\bar{\beta}$ in (31); in the MOM scheme one sets $l=5/6$ and replaces β by Ψ . The onset of the sign-alternating behaviour of the coefficients is delayed in the on-shell scheme by the presence of an unabsorbed large logarithm.

Tables 1 and 2 give a selection of our analytical and numerical results. Further results may be obtained by emailing the author. The possibility of using resummation techniques on the sign-alternating series of Table 3 is under study (in collaboration with A.L. Kataev).

The substantial increase in analytical QED results, reported here, was achieved at the expense of only a few hours of CPUtime. The extension to the next order in $1/N$ would be much more demanding, since there one encounters the light-by-light scattering contributions of [8, 17].

Acknowledgements. I am grateful to John Gracey for discussions of the large N expansion in the $\overline{\text{MS}}$ scheme [15, 16], which suggested the possibility of also obtaining on-shell results to all orders in the coupling. I thank Tony Hearn, for help with REDUCE,

and Andrei Kataev, Dmitri Kazakov, Anatoly Kotikov, Eduardo de Rafael, Ettore Remiddi and Oleg Tarasov, for discussion and advice. Hospitality in Marseille, Bologna, and Dubna is gratefully acknowledged. I am grateful for the support of SERC at the UK HEP Institute, in Durham, where this work was largely completed.

References

1. T. Kinoshita, H. Kawai, Y. Okamoto: Phys. Lett. B254 (1991) 235
2. R.N. Faustov, A.L. Kataev, S.A. Larin, V.V. Starshenko: Phys. Lett. B254 (1991) 241
3. A. Palanques-Mestre, P. Pascual: Comm. Math. Phys. 95 (1984) 277
4. D.J. Broadhurst, A.L. Kataev, O.V. Tarasov: Phys. Lett. B298 (1993) 445; D.J. Broadhurst: OUT-4102-40 (1992)
5. T. Kinoshita: CLNS 93/1187 (January 1993)
6. S.G. Gorishny, A.L. Kataev, S.A. Larin: Phys. Lett. B273 (1991) 141; B275 (1992) 512 (Erratum)
7. D.J. Broadhurst: Z. Phys. C54 (1992) 599
8. T. Kinoshita, B. Nizic, Y. Okamoto: Phys. Rev. D41 (1990) 593
9. A.L. Kataev: Phys. Lett. B284 (1992) 401
10. E. de Rafael, J.L. Rosner: Ann. Phys. 82 (1974) 369
11. H. Kawai, T. Kinoshita, Y. Okamoto: Phys. Lett. B260 (1991) 193
12. D.J. Broadhurst: Z. Phys. C32 (1986) 249
13. D.T. Barfoot, D.J. Broadhurst: Z. Phys. C41 (1988) 81
14. A.N. Vasil'ev, Yu.M. Pis'mak, Yu.R. Khonkonen: Theor. Math. Phys. 46 (1981) 104; 47 (1982) 465; 50 (1982) 127
15. J.A. Gracey: Phys. Lett. B246 (1990) 114; B262 (1991) 49; J. Phys. A23 (1990) 2183; J. Mod. Phys. A6 (1991) 395; Nucl. Phys. 348 (1991) 737; B352 (1991) 183
16. J.A. Gracey: J. Phys. A24 (1991) L431; Mod. Phys. Lett. A7 (1992) 1945
17. S.G. Gorishny, A.L. Kataev, S.A. Larin, L.R. Surguladze: Phys. Lett. B256 (1991) 81
18. R. Coquereaux: Phys. Rev. D23 (1981) 2276
19. J. Calmet, E. de Rafael: Phys. Lett. B56 (1975) 181
20. A.C. Hearn: REDUCE user's manual, version 3.4, Rand publication CP78 (1991)
21. B. Lautrup, E. de Rafael: Nucl. Phys. B70 (1974) 317; B78 (1974) 576 (Erratum)
22. R. Barbieri, E. Remiddi: Nucl. Phys. B90 (1975) 233